RePivot — Pivot Adjuster

Version 1.1.0 | Publisher: SplashArt Toolbox | Namespace: `io.splashart.RePivot`

Requirements

Requirement	Minimum Version
Unity	6 (6000.0) or newer
Render Pipeline	Any (Built-in, URP, HDRP)
Platform	Editor only — no runtime footprint

Dependencies: `com.unity.modules.uielements`, `com.unity.modules.imgui` (both included in Unity by default).

> **Note:** This tool uses UI Toolkit and Unity 6 APIs exclusively. It will **not** compile on Unity 2022 or earlier.

Installation

From the Unity Asset Store

1. Open **Window** → **Asset Store** (or the Unity Hub Asset Store tab).
2. Search for **"RePivot"** by SplashArt Toolbox.
3. Click **Import** and accept all files.
4. Verify the following folder exists:

```
Assets/RePivot/
```

From a `.unitypackage` File

1. Double-click the downloaded `.unitypackage` file.
2. In the Import dialog, ensure **all files** are selected.
3. Click **Import**.

How to Use

Opening the Tool

Navigate to:

```
Tools → SplashArt → Pivot Adjuster
```

The Pivot Adjuster window will open as a dockable panel.

Basic Workflow

1. **Select a GameObject** in the Scene or Hierarchy.
   - The window displays the object name and current status.
   - If the object is a Prefab instance, a warning and an **Unpack Prefab** button appear. Unpack before adjusting.
2. **Choose a pivot position** using one of three methods:

Method	Description
Quick Presets	Click one of the 9 grid buttons (Top-Left, Center, Bottom-Right, etc.) to snap the pivot to a bounding-box anchor.
Manual Entry	Type exact world-space coordinates into the Vector3 field.
Scene Handle	Drag the orange position handle directly in the Scene View.

3. **Click "Apply Pivot"**.
   - A wrapper GameObject (`ObjectName_Pivot`) is created at the chosen position.
   - The original object is re-parented under the wrapper.
   - The object does **not** visually move — only the pivot (parent origin) changes.
4. **Undo** at any time with **Ctrl+Z** / **Cmd+Z**. The entire operation is collapsed into a single undo step.

Re-adjusting a Pivot

If the target already has a `_Pivot` wrapper, clicking **Apply Pivot** repositions the existing wrapper instead of nesting a new one.

Scene View Visualisation

While the window is open and an object is selected:

- A **cyan wireframe cube** shows the combined renderer bounds.
- An **orange disc and position handle** marks the proposed pivot location.
- A **"Pivot" label** floats above the handle for clarity.

Known Limitations

- **UI elements (RectTransform) are not supported.** The tool detects RectTransform components and disables controls with a status message.
- **Non-uniform ancestor scale.** When a target's parent hierarchy contains non-uniform scale, the child's `lossyScale` restoration after re-parenting is approximate. This is a Unity engine limitation — `lossyScale` is read-only and cannot always be decomposed exactly.
- **Multi-selection.** When multiple GameObjects are selected, only the active (primary) object is shown and operated on.
- **Prefab instances must be unpacked** before adjusting. The tool provides a one-click Unpack button.

Troubleshooting

Problem	Solution
"No object selected" message	Select a GameObject in the Scene Hierarchy. Project assets are not supported.
"UI elements not supported"	The selected object has a RectTransform. Pivot Adjuster works with standard Transform objects only.
"Prefab instance — unpack before adjusting"	Click the **Unpack Prefab** button in the tool window.
Controls disabled during Play mode	Exit Play mode. Scene modifications are disabled while playing.
Bounds wireframe not visible	The object has no enabled Renderer components.
Pivot position shows NaN	Ensure the manual coordinates contain valid numbers.

API Reference

All core logic is in the `PivotLogic` static class (`io.splashart.RePivot` namespace).

Method	Signature	Description
`AdjustPivot`	`GameObject AdjustPivot(GameObject target, Vector3 newPivotWorldPos)`	Creates (or repositions) a `_Pivot` wrapper parent at the given world position. Returns the wrapper, or `null` if validation fails. Single undo step.
`CalculateBounds`	`Bounds CalculateBounds(GameObject target)`	Returns the combined world-space AABB of all enabled renderers (excluding `ParticleSystemRenderer`).
`CalculateAnchorPosition`	`Vector3 CalculateAnchorPosition(Bounds bounds, BoundsAnchor anchor)`	Maps a `BoundsAnchor` enum value to a world-space point on the given bounds.
`IsSafeToModify`	`bool IsSafeToModify(GameObject target)`	Returns `false` if the target is part of an unpacked Prefab instance.
`IsSceneObject`	`bool IsSceneObject(GameObject target)`	Returns `true` if the target exists in a scene (not a Project asset on disk).
`HasPivotWrapper`	`bool HasPivotWrapper(GameObject target)`	Returns `true` if the target's immediate parent ends with `_Pivot`.
`HasVisibleRenderers`	`bool HasVisibleRenderers(GameObject target)`	Returns `true` if the target or children have at least one enabled, non-particle renderer.
`IsUIElement`	`bool IsUIElement(GameObject target)`	Returns `true` if the target has a `RectTransform` component.

`IsFinite`	`bool IsFinite(Vector3 v)`	Returns `true` if all three components are finite.

Folder Structure

```
Assets/RePivot/
├── CHANGELOG.md
├── Third Party Notices.txt
├── RePivot Welcome.asset
├── Documentation/
│   ├── README.md
│   ├── LICENSE.md
│   ├── RePivot-Documentation.html
│   └── RePivot-Documentation.pdf
├── Editor/
│   ├── io.splashart.RePivot.Editor.asmdef
│   ├── PivotAdjusterCore.cs
│   ├── PivotAdjusterWindow.cs
│   ├── RPDocHelper.cs
│   ├── RPReadme.cs
│   ├── RPReadmeEditor.cs
│   └── Resources/
│       ├── PivotAdjusterWindow.uxml
│       └── PivotAdjusterWindow.uss
├── Samples/
│   └── RePivotDemo.unity
└── Tests/
    └── Editor/
        ├── io.splashart.RePivot.Tests.Editor.asmdef
        └── PivotLogicTests.cs
```

Frequently Asked Questions

Can I use this on Prefab instances?

You must unpack the Prefab first. The tool detects Prefab instances and offers a one-click Unpack button. Alternatively, open the Prefab in Prefab Mode to edit its internal hierarchy directly.

Does this add any runtime components or scripts?

No. The assembly definition is Editor-only. Nothing is included in player builds.

What happens to existing child objects?

They remain as children of the original object, which is now a child of the new wrapper. The entire sub-hierarchy is preserved.

Can I undo the pivot change?

Yes. Press Ctrl+Z (Cmd+Z on macOS) to undo. The wrapper is destroyed and the original hierarchy is restored.

Does the tool work with all render pipelines?

Yes. Pivot Adjuster manipulates transforms and hierarchy only — it does not interact with materials, shaders, or rendering features. It works identically on Built-in, URP, and HDRP.
